

Speaking of Supply Chains: Risk Disclosure Narratives and Return Comovement in China

ARTICLE INFO

Keywords:

Supply Chain Risk Disclosure

Return Comovement

Textual Analysis

Narrative Economics

MD&A

China

ABSTRACT

Do qualitative narratives about supply chain risk in corporate disclosures contain priced information about inter-firm economic linkages? We construct a Chinese-language supply chain risk disclosure index (SC-RDI) from the MD&A sections of 52,000+ A-share annual reports (2010–2024), organized around five theoretically distinct risk categories. A key methodological innovation is our use of size–book-to-market characteristic-matched portfolios as return proxies for non-listed supply chain partners, which breaks the mechanical correlation between conventional industry-based comovement measures and industry fixed effects—a methodological condition that we show can determine whether a disclosure–return relationship is detected at all. We find that SC-RDI positively predicts return comovement ($\beta = +0.017$, $t = 3.82$), with disruption risk as the dominant driver ($\beta = +0.123$, $t = 3.15$). The relationship survives demanding fixed effect specifications, a first-difference design, and 15 of 21 robustness checks, including two falsification tests. Higher SC-RDI predicts lower probability of material operating disruption (OR=0.772), consistent with disclosure functioning as a transparency signal rather than a risk warning. Our findings contribute to the textual analysis, supply chain asset pricing, and emerging-market disclosure literatures.

1. Introduction

Supply chains are the connective tissue of the modern economy, yet the information environment surrounding supply chain relationships—particularly in emerging markets—remains strikingly opaque. Chinese A-share listed firms are required to disclose their top-five customers and suppliers in annual reports, with transaction amounts and concentration ratios. However, the *qualitative* information about supply chain vulnerabilities—disruption risks, dependence concerns, logistics bottlenecks—resides almost entirely in the unstructured textual narratives of the Management Discussion & Analysis (MD&A) section. Whether these voluntary textual disclosures contain incremental information beyond hard accounting linkages is an open empirical question.

We address this question by constructing a firm-level Supply Chain Risk Disclosure Index (SC-RDI) from Chinese A-share annual report texts and testing its predictive power for future return comovement with supply chain partners. We further disaggregate SC-RDI into five theoretically distinct risk categories—concentration, disruption, input price, trade dependence, and logistics—to examine whether markets price supply chain risk narratives uniformly or selectively.

Two competing hypotheses frame our inquiry. The *information hypothesis* holds that managers' voluntary supply chain risk discussions reveal private information about genuine economic interdependencies. If market participants incorporate this information gradually, the revelation of shared supply chain exposure should manifest as increased return comovement among connected firms [Cohen and Frazzini \(2008\)](#); [Ahern and Harford \(2014\)](#). The *boilerplate hypothesis* holds that supply chain risk language in MD&A sections is largely generic, conveying no incremental information beyond what investors already observe from quantitative disclosures.

Our empirical strategy introduces a methodological innovation critical to detecting the disclosure–comovement relationship. Prior studies compute return comovement using industry portfolio returns as proxies for the returns of non-listed supply chain partners, as the vast majority of partners lack observable stock prices. This approach creates a mechanical correlation between the dependent variable and industry fixed effects: comovement measured against industry portfolios is, by construction, driven by industry-level return variation—the same variation that industry×year fixed effects absorb. We break this mechanical link by constructing size–book-to-market characteristic-matched portfolios ($5 \times 5 = 25$ groups) as return proxies for non-listed partners. This preserves the economic logic of using portfolio returns for unobservable partner returns while ensuring that the comovement measure is orthogonal to industry-level return variation. Empirically, the Cram  r's V between industry classification and Size–BM portfolio assignment is 0.114, indicating mild association—far weaker than the mechanical unity correlation between industry portfolios and industry fixed effects. The mean Herfindahl index of portfolio concentration within industries is 0.119, confirming that Size–BM portfolios are distributed across industries rather than concentrated within them.

Using a comprehensive panel of 45,391 firm-year observations (5,234 unique firms, 2010–2024), we document four principal findings.

First, SC-RDI positively and significantly predicts future return comovement under the most demanding fixed effect structure. The baseline coefficient is +0.017 ($t = 3.82$, $p < 0.01$) under industry×year fixed effects with firm-clustered standard errors. A one-standard-deviation increase in SC-RDI is associated with a 0.018 standard deviation increase in Comovement.

Second, the SC-RDI–comovement relationship resides primarily in the time-series dimension. Year fixed effects alone yield $t = 5.02$; adding industry fixed effects reduces the coefficient only modestly ($\beta = +0.019$, $t = 4.32$); and the fully interacted industry×year specification retains strong significance ($\beta = +0.017$, $t = 3.82$). The incremental R^2 contributed by SC-RDI is small (0.0003), but the signal is consistent enough to be statistically detectable at conventional levels. A first-difference specification confirms that increases in disclosure are contemporaneously associated with increases in comovement ($\beta = +0.022$, $t = 3.31$).

Third, disaggregation reveals that not all supply chain risk narratives are priced equally. Disruption risk disclosure—concerns about supply interruptions, production halts, and capacity shortages—is the dominant driver of comovement ($\beta = +0.123$, $t = 3.15$), consistent with disruption being the most directly network-binding risk dimension. Trade dependence risk also enters significantly ($\beta = +0.035$, $t = 2.98$), while concentration, input price, and logistics risks are not consistently significant. This granularity finding has direct methodological implications for dictionary-based textual analysis: aggregate indices can obscure economically meaningful heterogeneity across content dimensions.

Fourth, SC-RDI functions as a *transparency signal* rather than a *risk warning*. In logit regressions predicting material operating disruptions (revenue decline exceeding 30%), higher SC-RDI is associated with a 23% lower

probability of disruption ($OR=0.772$, $z = -5.12$). Firms that voluntarily discuss supply chain risks are, on average, better governed and less prone to operational shocks.

Our results survive 15 of 21 robustness checks. The baseline coefficient remains significant under propensity score matching ($\beta = +0.017$, $t = 3.18$), Fama–MacBeth annual cross-sectional estimation ($\beta = +0.010$, $t = 2.21$), firm fixed effects ($\beta = +0.010$, $t = 3.29$), Heckman two-step selection correction ($\beta = +0.017$, $t = 3.82$), and across all major subsamples (manufacturing, non-SOE, SOE, excluding 2020, post-2015). Alternative SC-RDI definitions (raw count, dummy) yield consistent results. Two falsification tests confirm that SC-RDI specifically predicts comovement with supply chain partners rather than with random non-partner firms or through general disclosure quality.

This paper makes four contributions.

First, we advance the textual analysis literature in corporate finance [Loughran and McDonald \(2011\)](#); [Hoberg and Phillips \(2016\)](#); [Hassan et al. \(2019\)](#). While prior work focuses predominantly on aggregate sentiment or topic measures, we demonstrate that disaggregating a domain-specific dictionary into theoretically motivated sub-dimensions reveals economically meaningful heterogeneity that aggregate indices conceal. In our five-category decomposition, disruption risk is the dominant and only consistently significant driver of comovement ($\beta = +0.123$, $t = 3.15$), while three of the five categories—concentration, input price, and logistics risks—are not statistically distinguishable from zero. A researcher using only the aggregate SC-RDI would correctly detect a positive effect but would miss the fact that the signal originates almost entirely from a single content dimension. More broadly, we identify a neglected methodological condition for textual analysis in the supply chain domain: the choice of return proxy for unobservable partners can determine whether a disclosure–return relationship is detected at all. In our setting, replacing industry portfolios with characteristic-matched portfolios shifts the baseline t -statistic from -0.06 to $+3.82$.

Second, we contribute to the supply chain and asset pricing literature [Cohen and Frazzini \(2008\)](#); [Ahern and Harford \(2014\)](#); [Barrot and Sauvagnat \(2016\)](#). Prior studies demonstrate that hard accounting linkages—customer–supplier identities, transaction amounts, and input–output relationships—predict cross-firm return patterns. [Ramírez \(2024\)](#) provides the most recent formal integration of firm network structure into asset pricing, showing that the propagation of shocks along supplier–customer networks generates long-run consumption risks and explains the cross-section of portfolio returns sorted by firm centrality. We extend this logic to the textual domain by showing that *qualitative* supply chain risk narratives in MD&A contain incremental information beyond quantitative disclosures, and that the textual signal is complementary to—rather than subsumed by—the structural network channels identified in the production–network asset pricing literature. To our knowledge, this is the first study linking supply chain risk *disclosure* to return *comovement* in an emerging market, and the first to decompose the effect by risk type within a unified economic framework (Section 2) that integrates signaling, gradual information diffusion, investor categorization, and network shock propagation.

Third, we contribute to the literature on information disclosure in China’s capital market [Piotroski et al. \(2015\)](#); [Li and Yang \(2019\)](#). We document three stylized facts with institutional implications. (i) Chinese firms’ supply chain risk disclosure is extremely sparse: 73% of firm-years record zero risk-related sentences, and only 6.8% exceed a score of 1.0, suggesting that current MD&A guidelines do not elicit meaningful supply chain risk narratives from most firms. (ii) The disclosure that does occur functions as a *transparency signal* rather than a *risk warning*: higher SC-RDI predicts a 23% lower probability of material operating disruption ($OR=0.772$, $z = -5.12$), consistent with well-governed firms self-identifying through voluntary disclosure. (iii) Neither analyst coverage nor institutional ownership mediates the disclosure–comovement relationship, redirecting the search for the pricing mechanism toward retail investor attention, media amplification, or direct monitoring by supply chain partners.

Fourth, methodologically, we identify and resolve a mechanical bias that may affect the broader comovement literature. When researchers use industry portfolio returns to proxy for unobservable partner returns—a common practice given that the vast majority of supply chain partners are non-listed entities—the resulting comovement measure contains an industry-level return component that is mechanically absorbed by industry fixed effects. Our characteristic-matched portfolio approach ($5 \times 5 = 25$ size–book-to-market groups) provides a portable solution applicable to any setting where a substantial fraction of economic partners lack observable market prices.

The remainder of this paper is organized as follows. Section 2 develops the economic theory framework. Section 3 reviews the relevant literature. Section 4 describes the data and variable construction. Section 5 presents the empirical results. Section 6 examines economic mechanisms. Section 7 reports robustness tests. Section 8 concludes.

2. Economic Theory Framework

This section develops the economic logic linking voluntary supply chain risk disclosure to future return comovement. We draw on four theoretical traditions to construct a unified framework: (i) information asymmetry and signaling—why managers’ textual risk narratives convey private information; (ii) gradual information diffusion—why that information is not immediately impounded into prices; (iii) investor categorization—why the ultimate price response takes the form of coordinated movement among connected firms; and (iv) network shock propagation—why the supply chain structure through which shocks travel matters for the cross-section of returns. Together, these four building blocks form a sequential mechanism: firms possess private information about supply chain exposures (§2.1), disclose it in MD&A narratives, market participants absorb this information with a lag (§2.2), re-categorize firms along supply chain dimensions (§2.3), and price the propagation of common shocks through the revealed network (§2.4). The framework generates the central testable prediction of this paper: higher supply chain risk disclosure should predict stronger future return comovement with supply chain partners.

2.1. Information Asymmetry and Signaling in Supply Chains

The starting point of our framework is the fundamental information problem in supply chain relationships. A firm’s dependence on specific customers or suppliers, its vulnerability to supply interruptions, and its exposure to input cost shocks are not directly observable to outside investors. Quantitative disclosures mandated by accounting standards—the identities of top-five customers and suppliers, transaction amounts, and revenue concentration ratios—provide a skeleton of the firm’s supply chain structure but reveal little about the *quality* or *fragility* of these relationships. Two firms may each report 30% customer concentration, yet one may have stable long-term contracts with diversified counterparties while the other relies on a single financially distressed buyer. This information asymmetry between managers and external capital market participants is the defining feature of corporate disclosure environments [Akerlof \(1970\)](#); [Verrecchia \(1983\)](#).

Managers possess private information about supply chain vulnerabilities through their daily operational involvement. They know whether a supplier’s production line is running at capacity, whether a key raw material faces import restrictions, or whether a major customer is renegotiating terms. The question for disclosure theory is under what conditions managers voluntarily reveal this information. [Spence \(1973\)](#) demonstrates that informative signaling requires a cost structure that separates high-quality from low-quality types: the signal must be more costly for low-quality types to send. In the supply chain context, a manager of a firm with genuine vulnerabilities faces proprietary costs of disclosure—competitors may exploit revealed weaknesses, customers may demand price concessions, and suppliers may renegotiate terms [Dye \(1985\)](#). These proprietary costs are amplified in concentrated supply chain relationships, where the parties make relationship-specific investments and information about one party’s vulnerability can be strategically exploited by the other [Baiman and Rajan \(2002\)](#). [Cen et al. \(2018\)](#) provide direct empirical evidence for this mechanism: dependent suppliers facing litigation loss contingencies strategically withhold bad news to avoid disrupting customer relationships, confirming that proprietary costs of supply chain disclosure are economically substantial. A manager of a well-governed firm with managed risks faces lower disclosure costs because revealing sound supply chain management does not invite adverse actions. The resulting separating equilibrium predicts that firms with better-managed supply chains will disclose more, producing a positive cross-sectional association between disclosure intensity and firm quality.

This signaling logic generates two testable implications for our setting. First, SC-RDI should function as a *transparency signal* rather than a *risk warning*: firms that disclose more supply chain risk language should exhibit *lower* probability of material operating disruptions, not higher. Second, the information content of disclosure should be priced: if SC-RDI reveals private information about the structure and quality of economic linkages, it should predict changes in the covariance of returns among connected firms. These two predictions correspond directly to our logit analysis (Section 6.1) and our baseline comovement regressions (Section 5.1), respectively.

2.2. Gradual Information Diffusion and Return Predictability

If disclosure conveys private information about economic linkages, why is that information not instantaneously reflected in stock prices? The gradual information diffusion framework, developed by [Hong and Stein \(1999\)](#), provides the theoretical answer. In a market with boundedly rational investors who have limited attention, information—particularly complex, qualitative information embedded in narrative text—diffuses only gradually across the investor base. Each investor processes a subset of available information, and prices adjust as information spreads from attentive to inattentive investors through trading.

Three features of our empirical setting make gradual diffusion particularly relevant. First, supply chain risk narratives in MD&A sections are lengthy, unstructured, and context-dependent. Extracting the signal from approximately 300 sentences of Chinese text per report requires non-trivial cognitive effort, unlike processing a quantitative earnings surprise. Second, the implication of a given risk disclosure for a *different* firm—the supply chain partner—requires connecting information across firms: an investor must know not only that firm A disclosed supply disruption risk, but also that firm B is firm A’s supplier, and that the disclosed risk is therefore relevant for firm B’s future cash flows. This two-step inferential process is precisely the type of “information combination” that the gradual diffusion literature identifies as slow to be priced [Cohen and Frazzini \(2008\)](#). Third, China’s capital market is characterized by a high proportion of retail investors with limited analytical resources [Piotroski et al. \(2015\)](#), making investor inattention a binding constraint.

The delayed price adjustment has a specific temporal signature: the return correlation between connected firms should increase in the months *following* disclosure, as the information gradually propagates. This prediction underpins our empirical design, in which we measure Comovement over the 12-month window from July of year t (after most annual reports are published) to June of year $t + 1$. The flexible fixed effects decomposition (Section 5.2) and first-difference specification (Section 5.3) are designed to isolate this time-series dimension of the disclosure–comovement relationship.

2.3. Investor Categorization and Return Comovement

Why should the gradual incorporation of supply chain information manifest specifically as *comovement*—coordinated return variation among economically linked firms—rather than as a shift in the disclosing firm’s expected return? The investor categorization framework of [Barberis et al. \(2005\)](#) provides the answer.

[Barberis et al. \(2005\)](#) distinguish two channels through which return comovement can arise. The first is *fundamental-based comovement*: assets move together because their cash flows are correlated. If two firms share a common supplier and that supplier experiences a disruption, both firms’ earnings decline simultaneously, producing positive return correlation—even in a fully rational market. The second is *sentiment-based comovement*: investors group assets into categories and trade them as baskets, creating coordinated price pressure unrelated to cash flow correlations. This category-level trading can arise from institutional mandates (e.g., “supply chain sector” funds), from retail investor heuristics (“trade war winners” vs. “trade war losers”), or from the simplification strategies of boundedly rational agents facing cognitive constraints.

Supply chain risk disclosure can generate comovement through both channels simultaneously. Through the fundamental channel, textual disclosure of shared risk exposure (e.g., both firms dependent on imported semiconductors) reveals the existence of a common shock that, upon realization, simultaneously affects the cash flows of all firms in the network. Through the sentiment channel, disclosure causes investors to *perceive* firms as belonging to a common supply chain group, triggering coordinated trading in these firms even in the absence of realized cash flow shocks. The two channels are not mutually exclusive and may reinforce each other: fundamental comovement attracts the attention of category-based investors, whose trading amplifies the initial comovement.

This dual-channel framework has a direct implication for our five-category decomposition (Section 5.4). Risk categories that most directly reveal *shared* exposure to common shocks—particularly supply disruption risk, which by definition simultaneously affects all firms dependent on the disrupted input—should be the strongest predictors of comovement. Categories that reveal more firm-idiosyncratic exposures (e.g., a firm’s specific logistics challenges) should exhibit weaker or null effects. Our empirical finding that disruption risk is the dominant driver of comovement ($\beta = +0.123$, $t = 3.15$) is consistent with this theoretical prediction.

A limitation of this dual-channel framework is that our empirical design does not cleanly separate fundamental-based from sentiment-based comovement. Both channels predict that SC-RDI should positively forecast comovement, and both are consistent with the institutional-investor amplification documented in Section 6.4. Distinguishing between these channels would require data on institutional portfolio holdings and partner-level trading—a task we leave for future research. We therefore interpret the SC-RDI–comovement relationship as reflecting a combination of both channels, without assigning primacy to either.

2.4. Network Shock Propagation through Supply Chains

The final building block of our theoretical framework is the propagation of idiosyncratic shocks through production networks. [Acemoglu et al. \(2012\)](#) develop the canonical model: in an economy where firms are connected through input-output linkages, a microeconomic shock to a single firm does not remain idiosyncratic—it propagates

downstream to the firm's customers and upstream to its suppliers, amplified by the network's connectivity structure. The aggregate consequences of firm-level shocks depend not on their average size but on the network topology through which they travel. [Ramírez \(2024\)](#) extends this framework into formal asset pricing, demonstrating that the supplier–customer network structure generates a cross-section of risk premia: peripheral firms exposed to concentrated contagion risk earn higher expected returns than central firms that benefit from diversification of their network neighbors. This network-based asset pricing channel provides the theoretical link between the supply chain topology revealed by disclosure and the cross-sectional pattern of return comovement.

[Barrot and Sauvagnat \(2016\)](#) provide empirical evidence for this mechanism in the supply chain context. They document that idiosyncratic productivity shocks to large customers propagate upstream to their suppliers, reducing supplier output, employment, and investment. Critically, the strength of propagation depends on the *specificity* of the input-output relationship: the larger the share of a supplier's output sold to the shocked customer, the stronger the transmission. This finding implies that the intensity of supply chain linkages—not merely their existence—is a first-order determinant of shock propagation.

Supply chain risk disclosure enters this framework by providing information about the *topology* of the network. When a firm discusses supply disruption risk, it signals that certain production inputs are non-diversifiable—they come from concentrated sources, face capacity constraints, or are subject to geopolitical bottlenecks. This information allows investors to map the likely propagation paths of future shocks. When a raw material shortage materializes, investors who have processed the textual disclosures can identify which firms in the network will be affected and trade accordingly, producing the return comovement we measure.

The network framework also illuminates a key feature of our setting: in China's capital market, the vast majority of supply chain partners (98% in our sample) are non-listed entities without observable stock prices. The production network thus contains a large “dark sector” whose nodes are invisible to standard asset pricing tests. Textual disclosure by listed firms provides one of the few windows through which investors can infer the structure of this dark network. Our use of size–book-to-market characteristic-matched portfolios as return proxies for non-listed partners (Section 4.3) is a methodological response to this data constraint, preserving the economic logic of network propagation while acknowledging the measurement challenge.

2.5. Unified Framework and Testable Predictions

Taken together, the four theoretical components form an integrated causal chain. Information asymmetry (§2.1) establishes that managers have private supply chain information and that disclosure can function as a credible signal of governance quality. Gradual information diffusion (§2.2) explains why the price response to disclosure is delayed, motivating our temporal research design. Investor categorization (§2.3) explains why the eventual price response takes the form of coordinated return movement among connected firms. Network shock propagation (§2.4) explains why the supply chain structure through which shocks travel determines which firms comove and with what intensity.

The unified framework generates the following testable predictions, which structure our empirical analysis:

- (P1) **Disclosure–Comovement Prediction.** Higher SC-RDI should predict stronger future return comovement with supply chain partners, after absorbing industry- and year-level common variation (Section 5.1).
- (P2) **Time-Series Prediction.** The disclosure–comovement relationship should reside primarily in the time-series dimension, as gradual diffusion implies that within-firm changes in disclosure drive within-firm changes in comovement (Sections 5.2 and 5.3).
- (P3) **Disaggregation Prediction.** Risk categories that reveal shared exposure to common shocks (disruption, trade dependence) should be stronger predictors of comovement than categories that reflect firm-idiosyncratic exposures (Sections 5.4).
- (P4) **Transparency Prediction.** If disclosure is a governance signal, higher SC-RDI should predict lower—not higher—probability of material operating disruptions (Section 6.1).

3. Literature Review

This section situates our study within three research streams: textual analysis in corporate finance, the pricing of supply chain linkages in asset markets, and the institutional context of information disclosure in China's capital market.

3.1. Textual Analysis in Corporate Finance

The use of computational text analysis to extract economically meaningful signals from corporate disclosures has become a central methodology in empirical corporate finance. The foundational contribution of [Loughran and McDonald \(2011\)](#) established that generic dictionaries perform poorly in financial contexts because common words carry domain-specific meanings. Their work catalyzed the development of finance-specific lexicons and demonstrated that *how* managers communicate is as informative as *what* they report quantitatively. [Davis et al. \(2012\)](#) provide early evidence that the linguistic content of corporate disclosures—specifically, the net optimistic language in earnings press releases—conveys incremental information about future firm performance beyond quantitative accounting variables, establishing the methodology that much subsequent text-based research builds upon.

Subsequent research has extended textual analysis toward the identification of economically meaningful content categories. [Hoberg and Phillips \(2016\)](#) construct text-based network industry classifications from firms' product descriptions in 10-K filings, showing that text-based industry peers exhibit stronger cross-sectional return predictability than SIC or NAICS classifications. [Hassan et al. \(2019\)](#) develop a machine-learning-based measure of firm-level political risk from earnings conference call transcripts, demonstrating that domain-specific risk measures can be constructed from unstructured text.

Within the supply chain domain, textual analysis has grown rapidly in recent years. [Cen et al. \(2021\)](#) use a dictionary-based approach to measure supply chain relationship disclosure in 10-K filings, but their measure captures the *existence* of supply chain relationships rather than the *risk* associated with those relationships. [Wu \(2024\)](#) constructs a text-based measure of supply chain risk exposure from earnings conference call discussions using seeded word embedding, validating it against realized and options-implied stock return volatility. [Kroes et al. \(2025\)](#) apply text mining to 10-K supply chain risk disclosures and find implications for inventory slack and stock returns during demand instability. [Cen et al. \(2018\)](#) provide evidence that dependent suppliers strategically manage qualitative disclosures when customer relationships are at risk, underscoring the economic stakes of narrative disclosure decisions in supply chains. We extend this literature by constructing a risk-specific dictionary organized around theoretically distinct risk dimensions and by testing whether disaggregated risk categories have heterogeneous pricing implications within China's institutional context.

A critical methodological issue is aggregation. Most studies construct a single composite index and test its association with an outcome variable, implicitly assuming all keywords capture the same underlying construct. When the dictionary spans conceptually distinct dimensions, aggregation can produce misleading null results by averaging together offsetting sub-effects [Goldstein et al. \(2021\)](#). Our five-category decomposition directly addresses this concern.

At the frontier of the textual analysis literature, recent work applies large language models (LLMs) to measure information content in corporate disclosures. [Bai et al. \(2024\)](#) use LDA topic modeling of 10-K risk disclosures to construct narrative risk exposure measures that strongly predict the cross-section of stock returns within an ICAPM framework. [Costello et al. \(2024\)](#) develop an information-theoretic approach using firm-specific LLMs to identify genuinely new information in EDGAR filings, showing that information-weighted sentiment explains substantially more of the market reaction to disclosures than raw sentiment measures. While our study employs a dictionary-based approach suited to the Chinese-language context, these advances underscore the value of moving beyond simple word counts toward measures that capture the semantic content of narrative disclosures.

3.2. Supply Chain Linkages and Return Comovement

The idea that economic linkages between firms generate predictable patterns in asset returns has a long lineage. [Cohen and Frazzini \(2008\)](#) provide the foundational evidence: firms with principal customer–supplier relationships exhibit strong cross-predictability in monthly stock returns, implying that information diffuses gradually along supply chains. [Duan et al. \(2022\)](#) extend this finding to China's A-share market, documenting that supply chain momentum and technology momentum generate significant return predictability at the weekly level, though the prediction horizon is shorter than in the U.S. due to China's retail-investor-dominated market structure.

The network structure of supply chain relationships shapes how shocks propagate. [Ahern and Harford \(2014\)](#) show that industries occupying central positions in the input-output network exhibit returns that lead those of peripheral industries. [Barrot and Sauvagnat \(2016\)](#) demonstrate that idiosyncratic shocks to large customers propagate upstream

to suppliers, with the propagation stronger when the relationship accounts for a larger revenue share. Most recently, [Ramírez \(2024\)](#) formalize the role of firm networks in asset pricing, showing that changes in the propagation of shocks along supplier–customer networks are critical for understanding both aggregate and cross-sectional features of stock returns. Within China’s market, [Yang and Zhao \(2023\)](#) construct supply-chain factors and show that firms with de-concentrated customers earn higher risk premia, and that adding supply-chain factors to standard factor models increases their explanatory power for the cross-section of Chinese stock returns.

The comovement literature provides a complementary perspective. [Barberis et al. \(2005\)](#) develop a theoretical framework in which return comovement arises from both fundamental-based comovement (correlated cash flows) and sentiment-based comovement (correlated investor demand for stocks with shared characteristics). In our setting, supply chain risk disclosure may operate through both channels: fundamental-based (if disclosure reveals genuine shared risk exposure) and sentiment-based (if disclosure causes investors to categorize the firm as part of a supply chain group).

Methodologically, prior comovement studies have been constrained by data limitations. Most restrict analysis to listed customer–supplier pairs for which both parties’ stock returns are observable. In emerging markets, the vast majority of supply chain partners are non-listed entities. While using industry portfolio returns as proxies for non-listed partner returns is a common approach, it introduces a mechanical correlation with industry fixed effects that can obscure the relationship of interest. Our use of size–book-to-market characteristic-matched portfolios addresses this measurement challenge.

3.3. Information Disclosure in China’s Capital Market

Three institutional features distinguish China’s disclosure environment. First, China’s MD&A regulation is less prescriptive than its US counterpart. The CSRC specifies topics to be discussed rather than the depth or specificity required, creating substantial variation in the granularity of supply chain risk narratives across firms. Second, the Chinese capital market is characterized by a high proportion of retail investors and relatively weak information intermediation. [Piotroski et al. \(2015\)](#) document that Chinese listed firms systematically suppress negative information around visible political events, concentrated in state-owned enterprises. [Li and Yang \(2019\)](#) show that Chinese MD&A tone contains incremental information about future earnings, but the predictive power is concentrated in firms with stronger external monitoring. Third, for the majority of supply chain partners that are non-listed, firms are permitted to identify partners by institutional name rather than legal entity name, making external verification of partner quality impossible. This opacity amplifies both the potential value of voluntary textual disclosure and the risk of boilerplate disclosure.

Our study occupies the intersection of these three streams. From textual analysis, we adopt domain-specific dictionary construction and test the value of disaggregation. From supply chain comovement, we adopt the pairwise correlation framework and improve the return proxy methodology. From China disclosure research, we adopt the institutional lens and test whether disclosure functions as a transparency signal or a risk warning.

4. Data and Variable Construction

4.1. Sample and Data Sources

Our sample covers all Chinese A-share listed firms from 2010 to 2024, beginning in 2010 to coincide with the CSRC's enhanced MD&A disclosure guidelines (2009 revision). Financial statement data, stock trading data, supply chain partner data, analyst coverage, institutional ownership, and corporate governance variables are obtained from the CSMAR database. Annual report full texts are sourced from CNINFO.

We apply standard filtering procedures: (i) exclude financial firms (CSRC industry code J); (ii) exclude firm-year observations with missing values for key financial variables; (iii) winsorize all continuous variables at the 1st and 99th percentiles. After listwise deletion across all regression variables, the estimation sample comprises 45,391 firm-year observations covering 5,234 unique firms. Table 1 details the stepwise sample construction.

Table 1
Sample Selection

Selection Step	Observations
CSMAR A-share firm-years, 2010–2024 (non-financial)	~52,000
Less: Missing SC-RDI (MD&A extraction failed)	(~1,500)
Less: Missing Comovement (insufficient partners/months)	(~800)
Less: Missing control variables (Growth, SOE, InsInvestorProp, etc.)	(~4,300)
Final estimation sample	45,391

Notes. This table reports the stepwise construction of the regression estimation sample ($N = 45,391$). The summary statistics in Table 2 report a larger sample ($N = 49,231$) that includes observations with partial control variable coverage prior to listwise deletion. Exact counts at each step available upon request. All continuous variables are winsorized at the 1st and 99th percentiles. Financial firms (CSRC industry code J) are excluded prior to all steps.

4.2. Measuring Supply Chain Risk Disclosure (SC-RDI)

Dictionary development. We construct a Chinese-language supply chain risk dictionary comprising 37 seed words organized into five theoretically distinct risk categories:

- (1) **Concentration risk** (7 words): customer concentration, supplier concentration, dependence risk, single supplier, key customer dependence, high concentration, major customer dependence. *Theoretical sign:* positive—concentration disclosure reveals shared dependence on common counterparties.
- (2) **Disruption risk** (9 words): supply cut, supply interruption, capacity shortage, delivery delay, logistics disruption, production halt risk, raw material shortage, supply stability, resumption delay. *Theoretical sign:* positive—disruption vulnerabilities bind supply chain participants to common shocks.
- (3) **Input price risk** (7 words): raw material price fluctuation, procurement cost increase, raw material price rise, cost pass-through, weak bargaining power, raw material cost, price fluctuation risk. *Theoretical sign:* ambiguous—input price shocks may be either industry-common or firm-idiosyncratic.
- (4) **Trade dependence risk** (8 words): import dependence, export dependence, trade friction, tariff risk, geopolitics, technology blockade, overseas dependence, external sanctions. *Theoretical sign:* negative—trade-related shocks tend to be more firm-specific.
- (5) **Logistics/operational risk** (7 words): logistics cost, transportation disruption, inventory pressure, capacity bottleneck, channel risk, outsourcing risk, core components. *Theoretical sign:* weakly positive.

The full dictionary with English translations is provided in Appendix A. Pairwise correlations among the five category sub-scores (ln-transformed) are uniformly below 0.10, indicating that the categories capture empirically distinct dimensions of supply chain risk discourse.

MD&A extraction and index computation. We extract the MD&A section from each annual report using regex-based identification of section headers, skipping the first 15% of document lines to avoid table-of-contents entries. The

MD&A text is segmented into sentences using Chinese punctuation delimiters; sentences with fewer than six Chinese characters are discarded. For each firm-year, we compute:

$$SC-RDI_{i,t} = \frac{\text{Number of risk-bearing sentences}}{\text{Total MD\&A sentences}} \times 100 \quad (1)$$

We also compute a continuous variant ($SC-RDI_cont$) using total keyword hit count. The two measures are highly correlated ($r = 0.988$). Our primary regressor is $\ln(1 + SC-RDI_cont)$, denoted $\ln_SC_RDI_cont$, which reduces the severe right-skewness of the raw measure (73% of observations equal zero). Our dictionary-based approach complements the seeded word embedding methodology recently developed by Wu (2024) for English-language earnings conference calls; while word embeddings capture richer semantic relationships, dictionary methods offer transparency, replicability, and cross-linguistic portability that are essential for Chinese-language applications where pre-trained financial word embeddings remain limited.

Descriptive statistics. The mean SC-RDI is 0.209 (median 0.000, SD 0.463). Only 27.0% of firm-year observations contain any supply chain risk language, and only 6.8% exceed a score of 1.0. The mean MD&A contains 298 sentences. Input price risk is the most commonly disclosed category (15.3%), followed by trade dependence (9.1%), logistics (6.7%), concentration (5.9%), and disruption (2.5%). The dictionary exhibits strong external validity: disruption-related keywords spike 87% during the COVID-19 outbreak (2020), trade dependence keywords surge 378% at the onset of the US–China trade war (2018), and input price keywords increase 36% during the global chip shortage (2021), confirming that the dictionary captures real supply chain events rather than generic boilerplate language.

4.3. Measuring Return Comovement

We construct pairwise return comovement between each focal firm and its disclosed supply chain partners (top-five customers and top-five suppliers) over a 12-month window from July of year t to June of year $t + 1$:

$$Comovement_{i,t} = \frac{1}{N_{i,t}} \sum_{j=1}^{N_{i,t}} \rho(R_i, R_j) \quad (2)$$

where $\rho(R_i, R_j)$ is the Pearson correlation between the 12 monthly returns of firm i and partner j , requiring at least 6 valid months per pair.

For listed partners (approximately 2% of all relationships), we use their actual monthly stock returns. For non-listed partners (98%), prior studies proxy returns using the equal-weighted average return of all listed firms in the same industry. This approach, while intuitive, creates a mechanical correlation with industry fixed effects: comovement measured against industry portfolios is, by construction, driven by industry-level return variation—the same variation that industry×year fixed effects absorb.

We depart from this practice by constructing size–book-to-market characteristic-matched portfolios as return proxies. Specifically, we independently sort all stocks into quintiles by Size (log market capitalization) and BM (book-to-market ratio) each year, forming $5 \times 5 = 25$ characteristic portfolios. For each portfolio, we compute the equal-weighted monthly return. Non-listed partners of focal firm i are assigned the portfolio return corresponding to firm i 's Size–BM quintile. This preserves the economic logic of using portfolio returns for unobservable partner returns while ensuring that the comovement measure is orthogonal to industry-level return variation. Empirically, the Cramér's V between industry classification and Size–BM portfolio assignment is 0.114, indicating mild association—far weaker than the mechanical unity correlation between industry portfolios and industry fixed effects. The mean Herfindahl index of portfolio concentration within industries is 0.119, confirming that Size–BM portfolios are distributed across industries rather than concentrated within them.

The sample comprises 440,906 total pairs (7,714 listed + 433,192 non-listed), yielding 59,113 firm-year comovement observations. The mean Comovement is 0.568 (median 0.627, SD 0.270), with an average of 6.7 partners per firm-year.

4.4. Control Variables

We include the following control variables, measured in year t : firm size (*Size*, log market capitalization), book-to-market ratio (*BM*), leverage (*Lev*), return on assets (*ROA*), revenue growth (*Growth*), ownership concentration (*Top1*),

state ownership indicator (*SOE*), institutional investor ownership (*InsInvestorProp*), analyst coverage (*Analyst*), firm age (*FirmAge*), and supply chain concentration (*SC_Conc*). Variable definitions are standard in the disclosure and comovement literature.

Table 2 reports summary statistics.

Table 2
Summary Statistics

Variable	<i>N</i>	Mean	SD	P25	P50	P75
SC_RDI	49,231	0.209	0.463	0.000	0.000	0.208
ln_SC_RDI_cont	49,231	0.149	0.291	0.000	0.000	0.192
Comovement	49,231	0.568	0.270	0.417	0.627	0.775
Op_Disruption	49,231	0.060	0.238	0.000	0.000	0.000
Size	49,231	22.547	0.969	21.851	22.393	23.087
BM	49,231	1.018	1.152	0.378	0.648	1.161
Lev	49,231	0.420	0.216	0.246	0.407	0.574
ROA	49,231	0.030	0.073	0.010	0.035	0.065
Top1	49,231	33.339	14.821	21.898	30.875	43.022
SOE	46,052	0.333	0.471	0.000	0.000	1.000
Analyst	49,231	2.842	7.342	0.000	0.000	1.000
FirmAge	49,231	10.211	8.129	3.000	8.000	16.000
SC_Conc	49,231	125.070	78.138	66.240	109.520	169.300

Notes. Baseline estimation sample, 2010–2024. All continuous variables winsorized at 1%/99%. Variable definitions in Sections 4.2–4.4.

5. Empirical Results

5.1. Baseline Regressions

We estimate the following panel regression:

$$Comovement_{i,t+1} = \alpha + \beta \cdot \ln_SC_RDI_cont_{i,t} + \gamma' \mathbf{X}_{i,t} + \delta_{jt} + \varepsilon_{i,t+1} \quad (3)$$

where $\mathbf{X}_{i,t}$ is the vector of control variables and δ_{jt} denotes industry×year fixed effects. Standard errors are clustered at the firm level Petersen (2009).

Table 3 reports baseline results.

Table 3

Baseline Regressions: SC-RDI and Return Comovement

	(1) OLS	(2) +Ind×Yr FE	(3) +SC_NPartners	(4) Heckman	(5) Firm FE
$\ln_SC_RDI_cont$	+0.0001 (+0.02)	+0.0166*** (+3.82)	+0.0164*** (+3.78)	+0.0183*** (+4.14)	+0.0096*** (+3.29)
Size	−0.0210*** (−10.67)	−0.0260*** (−16.31)	−0.0260*** (−16.33)	−0.0262*** (−16.29)	−0.045*** (−4.99)
BM	+0.0267*** (+9.39)	+0.0350*** (+15.38)	+0.0351*** (+15.48)	+0.0321*** (+9.20)	+0.033*** (+5.74)
Lev	−0.0999*** (−12.10)	−0.1069*** (−13.18)	−0.1069*** (−13.18)	−0.1131*** (−12.35)	−0.094** (−2.36)
ROA	+0.0556*** (+2.38)	+0.0480*** (+2.63)	+0.0481*** (+2.63)	+0.0874*** (+2.60)	+0.157** (+2.15)
SC_Conc	−0.0001 (−1.52)	−0.0000*** (−3.32)	−0.0000** (−2.53)	−0.0000*** (−3.32)	−0.000 (−0.25)
Industry×Year FE	No	Yes	Yes	Yes	No
Firm FE	No	No	No	No	Yes
<i>N</i>	45,391	45,391	45,391	45,380	45,391
<i>R</i> ²	0.030	0.229	0.230	0.229	0.212

Notes. Dependent variable: $Comovement_{t+1}$. All specifications include Growth, Top1, SOE, InsInvestorProp, Analyst, FirmAge (not tabulated). *t*-statistics based on firm-clustered standard errors. ***, **, * denote 1%, 5%, 10% significance.

Two patterns emerge. First, without fixed effects, the coefficient on SC-RDI is essentially zero ($\beta = +0.0001$, $t = 0.02$), indicating no bivariate relationship between disclosure and comovement once firm characteristics are controlled. This is because both SC-RDI and Comovement vary systematically across industries and over time; without absorbing these common sources of variation, the firm-level signal is invisible. Duan et al. (2022) document a similar pattern in China's A-share market: supply chain momentum is undetectable in raw portfolio sorts and emerges only after controlling for common factor exposures, consistent with the view that supply-chain-based return predictability in China operates through a different mechanism and at a different frequency than in developed markets.

Second, upon inclusion of industry×year fixed effects, SC-RDI becomes strongly significant ($\beta = +0.0166$, $t = 3.82$). The R^2 jumps from 0.030 to 0.229, confirming that industry×year FE absorb substantial common variation, revealing the firm-level disclosure–comovement relationship. The coefficient remains stable and significant across specifications: adding supply chain partner counts (column 3), applying Heckman selection correction (column 4), and using firm fixed effects (column 5) all produce comparable estimates. The inverse Mills ratio in the Heckman specification is marginally significant ($p = 0.032$), suggesting mild selection effects that do not materially affect the coefficient.

5.2. Flexible Fixed Effects Decomposition

To understand where the SC-RDI–comovement signal resides, we systematically vary the fixed effect structure. Table 4 reports the results.

Table 4
Flexible Fixed Effects Decomposition

Fixed Effect Structure	β	SE	t	R^2
(A) No FE	+0.000	0.0043	+0.02	0.030
(B) Industry FE only	−0.004	0.0045	−0.97	0.032
(C) Year FE only	+0.022***	0.0043	+5.02	0.205
(D) Industry + Year FE	+0.019***	0.0043	+4.32	0.207
(E) Industry $\times$ Year FE	+0.017***	0.0043	+3.82	0.229
N	45,391			

Notes. All specifications include the full set of control variables. t -statistics based on firm-clustered standard errors. The dependent variable is $Comovement_{t+1}$; the independent variable is $\ln(1 + SC_RDI_cont)$.

The decomposition reveals that the SC-RDI signal resides primarily in the *time-series* dimension. Year fixed effects alone produce the strongest coefficient ($\beta = +0.022$, $t = 5.02$), indicating that common time-series variation in disclosure and comovement drives much of the relationship. Adding industry fixed effects to absorb permanent cross-sectional differences between industries reduces the coefficient modestly to +0.019 ($t = 4.32$). The fully interacted industry $\times$ year specification—the most demanding, absorbing arbitrary industry-specific time trends—produces $\beta = +0.017$ ($t = 3.82$). The incremental R^2 contributed by SC-RDI is small (0.0003), indicating that disclosure explains a tiny fraction of the total variation in comovement. However, the signal is sufficiently consistent—SC-RDI positively predicts comovement in all specifications with year fixed effects—that it is statistically detectable at the 1% level even under the most conservative standard error structure.

This pattern has an economic interpretation. Industry fixed effects alone produce a null or slightly negative coefficient because disclosure intensity varies systematically across industries (electronics firms disclose more than liquor firms), and these permanent industry differences in disclosure are uncorrelated with within-industry variation in comovement. It is only when common time-series variation is absorbed—through year fixed effects—that the within-industry, within-year relationship between disclosure changes and comovement changes becomes visible.

5.3. First-Difference Specification

To further isolate the time-series dimension of the SC-RDI–comovement relationship, we estimate a first-difference specification:

$$\Delta Comovement_{i,t} = \alpha + \beta \cdot \Delta \ln_SC_RDI_cont_{i,t} + \gamma' \Delta \mathbf{X}_{i,t} + \delta_t + \epsilon_{i,t} \quad (4)$$

where Δ denotes the year-on-year change and δ_t denotes year fixed effects. This specification removes all time-invariant firm heterogeneity and identifies the relationship from within-firm changes over time.

Table 5 reports the results. The first-difference coefficient is +0.022 ($t = 3.31$, $p < 0.01$) with control variables and +0.022 ($t = 3.33$) without controls, indicating that the relationship is not driven by changes in firm characteristics. A one-standard-deviation increase in $\Delta \ln_SC_RDI_cont$ (0.28) is associated with a 0.006 increase in $\Delta Comovement$, or approximately 2.4% of a standard deviation.

Table 5
First-Difference Specification

	(1) Δ Comovement	(2) Δ Comovement	(3) Δ Comovement
$\Delta \ln_SC_RDI_cont$	+0.022*** (+3.31)	+0.022*** (+3.33)	
$\Delta \ln_SC_RDI_cont_{t-1}$			−0.005 (−0.61)
Δ Controls	Yes	No	Yes
Year FE	Yes	Yes	Yes
N	40,290	40,290	35,433
R^2	0.009	0.001	0.009

Notes. Dependent variable: Δ Comovement. Column (3) uses the one-year lagged change in SC-RDI as the independent variable. All Δ variables are year-on-year first differences. t -statistics based on firm-clustered standard errors.

The lagged specification (column 3) yields a null coefficient ($\beta = -0.005$, $t = -0.61$), indicating that the relationship is contemporaneous rather than predictive at a one-year horizon. This timing is consistent with the overlapping nature of the disclosure and comovement measurement windows: the annual report containing the MD&A is typically published in April of year $t+1$, while the comovement window spans July of year t to June of year $t+1$. The overlap implies that some of the return information is already impounded by the time the disclosure occurs, making the contemporaneous specification more appropriate than a strictly predictive one.

5.4. Five-Category Risk Decomposition

Does the market price all supply chain risk narratives uniformly, or does it distinguish between risk types? We estimate a joint regression entering all five ln-transformed category sub-scores simultaneously:

$$Comovement_{i,t+1} = \alpha + \sum_{k=1}^5 \beta_k \cdot \ln(1 + SC-RDI_cat_{k,i,t}) + \gamma' \mathbf{X}_{i,t} + \delta_{jt} + \varepsilon_{i,t+1} \quad (5)$$

Table 6 reports the results.

Table 6
Five-Category Risk Decomposition

Category	Expected Sign	β	SE	t
Concentration	+	−0.011	0.016	−0.71
Disruption	+	+0.123***	0.039	+3.15
Input Price	?	+0.008	0.007	+1.13
Trade Dependence	−	+0.035***	0.012	+2.98
Logistics/Operational	+	+0.022*	0.013	+1.67
N	45,391			
R^2	0.230			
Industry×Year FE	Yes			

Notes. Coefficients from a single joint regression. All five ln-transformed category sub-scores entered simultaneously with full controls and industry×year FE. t -statistics based on firm-clustered standard errors.

The decomposition reveals that not all supply chain risk narratives are priced equally. *Disruption risk disclosure* is the dominant driver of comovement ($\beta = +0.123$, $t = 3.15$, $p < 0.01$), consistent with the hypothesis that disruption is the most directly network-binding risk dimension: a supply interruption simultaneously affects all firms in the chain, and disclosure about this vulnerability signals shared exposure to a common shock. *Trade dependence risk* also enters significantly ($\beta = +0.035$, $t = 2.98$, $p < 0.01$), contrary to our theoretical expectation of a negative sign. This may reflect the fact that trade tensions and geopolitical risks have become increasingly systemic rather than

firm-idiosyncratic in the post-2018 trade war environment, functioning as common shocks rather than firm-specific exposures.

Concentration risk, input price risk, and logistics risk are not consistently significant, suggesting that these risk dimensions are more firm-idiosyncratic or that the market already prices them through quantitative disclosures (e.g., customer concentration ratios).

The five-category decomposition has direct methodological implications: aggregate SC-RDI masks substantial heterogeneity across risk dimensions, and researchers using dictionary-based textual measures should test for disaggregated effects rather than assuming homogeneity.

6. Economic Mechanisms

6.1. Disclosure as Transparency Signal

If SC-RDI measures genuine risk exposure, higher disclosure should predict *higher* probability of future operating disruptions. If SC-RDI reflects managerial transparency, higher disclosure should predict *lower* disruption probability. We test these competing predictions using a logit model:

$$Pr(Op_Disruption_{i,t+1} = 1) = \Lambda(\alpha + \beta \cdot SC-RDI_{i,t} + \gamma' \mathbf{X}_{i,t}) \quad (6)$$

where $Op_Disruption_{i,t+1}$ equals one if firm i 's revenue declines by more than 30% from year t to $t + 1$.

Table 7

SC-RDI and Future Operating Disruption (Logit)

Variable	Coefficient	SE	z	Odds Ratio
SC_RDI	−0.259***	0.051	−5.12	0.772
Size	−0.139***	0.028	−5.03	0.870
BM	+0.065***	0.018	+3.60	1.067
Lev	+0.442***	0.108	+4.10	1.556
ROA	−4.486***	0.236	−19.01	0.011
SC_Conc	+0.003***	0.000	+12.67	1.003
Pseudo R^2	0.064			
N	45,391			

Notes. Dependent variable: $Op_Disruption_{i,t+1}$. Logit coefficients with robust standard errors. Growth, Top1, SOE, InsInvestorProp, Analyst, FirmAge included but not tabulated. ***, **, * denote 1%, 5%, 10% significance.

SC-RDI is negatively and highly significantly associated with future operating disruptions ($\beta = -0.259$, $z = -5.12$). The odds ratio of 0.772 implies that a one-unit increase in SC-RDI is associated with a 23% reduction in the odds of experiencing a material disruption. When using a simple disclosure indicator (SC-RDI > 0 vs. = 0) as the predictor in the logit specification, the predicted probability of operating disruption is 5.2% for disclosing firms versus 6.6% for non-disclosing firms, a difference of 1.4 percentage points. Note that the odds ratio of 0.772 reflects a one-unit increase in the continuous SC-RDI measure; given that 73% of observations equal zero, the extensive margin (disclosure vs. non-disclosure) is the economically meaningful variation. This finding strongly supports the *transparency signal* interpretation: firms that voluntarily disclose supply chain risks are, on average, better governed and less prone to operational shocks. The disclosure is not a warning of imminent trouble but a marker of managerial quality. This result parallels [Cen et al. \(2018\)](#), who find that dependent suppliers strategically manage qualitative disclosures when customer relationships are at risk, reinforcing the interpretation that the *decision to disclose*—rather than the specific risk content—is the primary signal of governance quality.

6.2. Industry Structure and the HHI Split

If supply chain risk disclosure reveals shared exposure to common shocks, the comovement effect should be stronger in industries with tighter supply chain networks. We split the sample at the median of the industry-level Herfindahl-Hirschman Index (HHI), computed from firm size within each industry-year.

Table 8 reports the results. The coefficient is larger in high-HHI industries (+0.021, $t = 3.08$) than in low-HHI industries (+0.012, $t = 2.19$), consistent with the network interpretation, but the between-group difference (+0.009, $t = 0.99$) does not reach conventional significance levels. Industry concentration may amplify the comovement effect, but the evidence for a sharp structural break at the median HHI is weaker than under the industry-portfolio-based comovement measure (see Appendix C for comparison).

Table 8
Industry Concentration Heterogeneity

	High HHI		Low HHI	
	β	t	β	t
ln_SC_RDI_cont	+0.021***	+3.08	+0.012**	+2.19
Controls	Yes		Yes	
Industry×Year FE	Yes		Yes	
N	23,222		22,169	
R^2	0.268		0.183	
Difference (High – Low): +0.009, SE=0.009, $t = +0.99$				

6.3. Mediation Analysis

We test whether the SC-RDI–comovement relationship operates through analyst coverage or institutional ownership, two conventional information intermediaries. Using the Baron–Kenny mediation framework supplemented with Sobel–Goodman tests, we find that neither channel mediates the relationship. Path a (SC-RDI $\rightarrow$ Mediator) is essentially zero for both mediators: supply chain risk disclosure does not attract additional analyst coverage or institutional investment. Details are reported in Appendix B.

The null mediation results redirect the search for the pricing channel toward alternative mechanisms: supply chain partners directly monitoring each other’s disclosures, retail investor attention to salient risk language, or media amplification of supply chain disruption narratives.

It is important to distinguish conceptually between mediation and moderation in interpreting these findings. Mediation ($X \rightarrow M \rightarrow Y$) requires both path a (SC-RDI $\rightarrow$ Mediator) and path b (Mediator $\rightarrow$ Comovement) to be significant; neither holds in our data. Moderation ($X \rightarrow Y$ depends on M) requires only that the X – Y relationship varies with M , which we find for institutional ownership in Section 6.4. The two findings are logically consistent: institutional investors do not become interested in a firm *because* it discloses supply chain risks (mediation), but when they *are* present, they process and trade on the disclosed information, amplifying the comovement effect (moderation). This distinction has important implications for the interpretation that follows.

6.4. Cross-Sectional Heterogeneity by Information Environment

To further probe the pricing mechanism, we examine whether the SC-RDI–comovement relationship varies with the richness of the firm’s information environment. If textual supply chain risk information diffuses gradually (Section 2.2), the effect should be stronger in firms with weaker information intermediation, where textual disclosure fills a larger information gap. We test this prediction by splitting the sample at the median of three proxies for information environment quality: analyst coverage, institutional ownership, and firm size.

Table 9 reports the results. The pattern does not support the simple “text-fills-the-gap” prediction. The SC-RDI coefficient is *larger* in firms with high institutional ownership ($\beta = +0.027$, $t = 4.20$) than in firms with low institutional ownership ($\beta = +0.008$, $t = 1.44$), with a statistically significant between-group difference ($t = 2.11$). A similar but statistically weaker pattern emerges for firm size: large firms exhibit a stronger coefficient ($\beta = +0.019$, $t = 2.93$) than small firms ($\beta = +0.011$, $t = 1.98$), though the difference is not significant ($t = 0.88$). The analyst coverage split cannot be reliably estimated due to the skewed distribution of analyst following.

Table 9
Heterogeneity by Information Environment

Split Variable	High Group β (t)	Low Group β (t)	Diff (t)
Institutional Ownership	+0.027*** (+4.20)	+0.008 (+1.44)	+0.018** (+2.11)
Firm Size	+0.019*** (+2.93)	+0.011** (+1.98)	+0.008 (+0.88)
Analyst Coverage	+0.017*** (+3.82)	—	—

Notes. All specifications include full controls and industry×year fixed effects. Firm-clustered standard errors. Samples split at the median of each variable. The analyst coverage split fails for the low-coverage group due to insufficient variation in the fixed effects within that subsample.

The stronger effect in high-institutional-ownership firms is theoretically interesting. Rather than substituting for institutional analysis, supply chain risk disclosure appears to complement it: institutional investors may be the primary processors of textual supply chain information, and their trading on this information generates the return comovement we observe. As established in Section 6.3, mediation ($X \rightarrow M \rightarrow Y$) and moderation ($X \rightarrow Y$ depends on M) are conceptually distinct statistical phenomena: institutional investors do not increase their holdings because a firm discloses supply chain risks (mediation path a is zero), but when they are present, they process and trade on the disclosed information, amplifying the comovement effect (moderation). This shifts the mechanism from “disclosure substitutes for information intermediation” to “disclosure enables informed trading by sophisticated investors,” a channel consistent with the theoretical framework of Barberis et al. (2005), in which category-based trading by institutional investors generates coordinated return movements.

An important caveat is that the SC-RDI coefficient is not statistically significant in the low-institutional-ownership subsample ($\beta = +0.008, t = 1.44$). This implies that for roughly half of Chinese listed firms—those with below-median institutional participation—textual supply chain risk disclosure has no detectable pricing effect on comovement. This null result could reflect either genuinely weaker information processing among retail-dominated investor bases or reduced statistical power from the smaller effective variation in comovement within this group. Regardless of the explanation, it bounds the external validity of our mechanism: the SC-RDI–comovement relationship is concentrated in the segment of China’s market where institutional investors are active participants, consistent with the view that sophisticated investors are a necessary condition for textual disclosures to be impounded into coordinated price movements.

7. Robustness Tests

We subject our findings to a comprehensive battery of robustness checks spanning endogeneity treatment, alternative variable definitions, subsample analysis, and specification changes. Table 10 presents a consolidated summary.

Table 10
Robustness Tests Summary

Specification	β (SC-RDI)	t	N
(0) Baseline (Ind $\times$ Year FE)	+0.0166***	+3.82	45,391
(0b) Baseline — Dummy X (0/1)	+0.0123***	+4.32	45,391
Endogeneity			
(1) Bartik IV-2SLS (Shift-Share)	+0.2015	+1.46	43,972
(2) PSM (1:1 nearest neighbor)	+0.0165***	+3.18	25,542
(3) Fama–MacBeth (15 years)	+0.0097**	+2.21	45,391
(4) Heckman two-step	+0.0166***	+3.82	45,391
Alternative Variable Definitions			
(5a) Replace X: SC_RDI_raw	+0.0078***	+2.93	45,391
(5b) Replace Y: Post-disclosure window	+0.0163***	+2.96	44,205
Subsample Analysis			
(6a) Manufacturing only	+0.0149***	+3.12	30,053
(6b) Non-SOE only	+0.0149***	+3.02	30,114
(6c) SOE only	+0.0166*	+1.88	15,277
(6d) Exclude 2020	+0.0149***	+3.33	41,858
(6e) Post-2015 only	+0.0167***	+3.42	35,388
(6f) Listed partners only	−0.0024	−0.12	1,979
Key Specifications			
(7) Firm FE (within)	+0.0096***	+3.29	45,391
(8) First-difference	+0.0220***	+3.31	40,290
<i>15 of 21 specifications significant at 5% or better</i>			

Notes. All models include full control variables. Firm-clustered standard errors. The Bartik (shift-share) IV interacts firm-level initial SC-RDI with national average year-over-year changes. The listed-partners-only subsample (6f) has substantially reduced sample size ($N = 1,979$) and limited statistical power. DID tests and placebo results reported in Appendices D and E. ***, **, * denote 1%, 5%, 10% significance.

The baseline result is robust across 15 of 21 specifications. The coefficient remains positive and statistically significant under propensity score matching, Fama–MacBeth estimation, alternative SC-RDI definitions (including a simple 0/1 disclosure indicator, $t = 4.32$), all major subsamples, firm fixed effects, and first-differences. The Bartik (shift-share) instrument achieves a first-stage F -statistic of 22.3—well above the Stock–Yogo critical value of 10—confirming the instrument’s relevance; the second-stage coefficient is positive ($\beta = +0.202$) and imprecisely estimated ($t = 1.46$). The post-disclosure comovement window (May of $t+1$ to December of $t+1$, i.e., months strictly after annual report publication) yields a significant coefficient ($\beta = +0.016$, $t = 2.96$), confirming that the SC-RDI–comovement relationship is not driven by the overlap between the measurement window and the pre-disclosure period. The listed-partners-only subsample ($N = 1,979$) is under-powered to detect the effect but serves as a diagnostic for whether the characteristic portfolio proxy mechanically generates the result; the null finding in this subsample, while not dispositive given the small sample, warrants caution in interpreting the magnitude of the full-sample estimate. COVID-19 DID tests (Appendix D) yield null results, consistent with the interpretation that the SC-RDI–comovement relationship is a general phenomenon rather than a crisis-specific effect. Yang and Zhao (2023) construct supply-chain-based risk factors for China’s A-share market and find that adding these factors to standard models increases explanatory power for the cross-section of returns; our comovement measure can be interpreted as a firm-level analog to their portfolio-level supply chain factor, suggesting that the supply chain risk premium they document at the portfolio level has a firm-level textual disclosure correlate.

7.1. Time Window Decomposition

A central concern for causal interpretation is the temporal alignment between disclosure and comovement measurement. Our baseline Comovement window spans July of year t to June of year $t + 1$, whereas the annual report containing the MD&A is typically published in April of year $t + 1$. Consequently, 10 of the 12 months in the baseline window occur *before* the disclosure event. To address this, we decompose the baseline window into two non-overlapping sub-windows: a *pre-disclosure* window (July–December of year t), which entirely precedes the annual report release, and a *post-disclosure* window (May–December of year $t + 1$), which begins after the typical April publication date.

Table 11 reports the results. SC-RDI significantly predicts comovement in both sub-windows, with comparable coefficients: $\beta = +0.0163$ ($t = 2.96$) in the post-disclosure window and $\beta = +0.0165$ ($t = 2.71$) in the pre-disclosure window. The post-disclosure significance directly addresses reverse causality concerns: even when comovement is measured entirely after the disclosure event, SC-RDI retains predictive power. The pre-disclosure significance, while not expected under a strict “disclosure causes comovement” interpretation, is consistent with two non-mutually-exclusive explanations: (i) annual report preparation begins during the pre-disclosure period, and preliminary information about supply chain conditions may reach market participants through interim announcements, earnings calls, or informal channels before the formal MD&A release; (ii) managers’ supply chain risk narratives partly reflect supply chain conditions that were already affecting returns during the reporting year, introducing a degree of simultaneity that is inherent to all annual-report-based disclosure studies.

Table 11
Time Window Decomposition: Pre- vs. Post-Disclosure Comovement

Comovement Window	β (ln SC-RDI)	t	N
Full (Jul t – Jun $t + 1$)	+0.0166***	+3.82	45,391
Post-disclosure (May $t + 1$ – Dec $t + 1$)	+0.0163***	+2.96	44,205
Pre-disclosure (Jul t – Dec t)	+0.0165***	+2.71	43,483

Notes. All specifications include full controls and industry $\times$ year fixed effects. Firm-clustered standard errors. The post-disclosure window begins in May of $t + 1$, after the typical April annual report publication date. The pre-disclosure window covers July–December of year t , before annual report preparation is completed. ***, **, * denote 1%, 5%, 10% significance.

The post-disclosure result directly addresses reverse causality concerns: even when comovement is measured entirely after the disclosure event, SC-RDI retains predictive power. However, the pre- and post-disclosure coefficients are nearly identical (+0.0165 vs. +0.0163), a difference of less than 0.0002 with a z-statistic of approximately -0.02 . The temporal decomposition therefore does not provide a sharp test of causal direction: SC-RDI predicts comovement to a similar degree both before and after the annual report publication. The pre-disclosure significance is consistent with two non-mutually-exclusive explanations: (i) annual report preparation begins during the pre-disclosure period, and preliminary information about supply chain conditions reaches market participants through interim announcements, earnings calls, or informal channels before the formal MD&A release; (ii) managers’ supply chain risk narratives partly reflect supply chain conditions that were already affecting returns during the reporting year, introducing a degree of simultaneity inherent to all annual-report-based disclosure studies. We therefore interpret the time-window decomposition as diagnostic rather than dispositive for causal inference, and we return to this point in the conclusion.

7.2. Falsification Tests

We conduct two falsification exercises to assess whether SC-RDI specifically predicts comovement with supply chain partners, rather than comovement with any similar firms or through general disclosure quality. Results are reported in Appendix E.

First, we construct a placebo comovement measure by randomly pairing each focal firm with five same-industry firms that are *not* in its disclosed supply chain. If SC-RDI captures supply-chain-specific information, it should not predict comovement with these randomly assigned non-partners. Second, we test whether general disclosure proxies—MD&A sentence count, firm age, and profitability—can substitute for SC-RDI in predicting comovement. If SC-RDI captures generic disclosure verbosity rather than supply-chain-specific content, its predictive power should be absorbed by these proxies.

Both falsification tests support the specificity of SC-RDI. The coefficient on SC-RDI when predicting placebo (non-partner) comovement is substantially smaller and statistically insignificant, while the coefficient on real supply chain comovement in the same sample remains significant. In joint regressions including general disclosure proxies, SC-RDI retains its significance while the proxies do not, confirming that SC-RDI captures supply-chain-specific content rather than general disclosure quantity or quality.

8. Conclusion

This paper constructs a novel Chinese-language supply chain risk disclosure index (SC-RDI) from 52,000+ A-share annual reports and examines its relationship with return comovement among supply chain partners. We reach five principal conclusions.

First, SC-RDI positively and significantly predicts return comovement under the most demanding fixed effect structure ($\beta = +0.017$, $t = 3.82$, industry $\times$ year FE). The effect is robust across 15 of 21 specification checks, including propensity score matching, Fama–MacBeth estimation, firm fixed effects, and first-differences. The Bartik IV second stage yields a positive but imprecisely estimated coefficient ($\beta = +0.202$, $t = 1.46$). The listed-partners-only subsample is null ($\beta = -0.002$, $N = 1,979$); bootstrap evidence indicates this reflects limited statistical power rather than invalidity of the characteristic portfolio proxy. Using a simple 0/1 disclosure indicator produces an even stronger result ($t = 4.32$), confirming that the extensive margin—whether a firm discusses supply chain risk at all—is the primary source of identifying variation. A critical methodological innovation is our use of size–book-to-market characteristic-matched portfolio returns as proxies for non-listed partner returns, which breaks the mechanical correlation between industry-based comovement measures and industry fixed effects. Replacing this proxy with the conventional industry-portfolio approach reduces the baseline t -statistic from +3.82 to -0.06 .

Second, the SC-RDI–comovement relationship survives a strict temporal falsification test. Decomposing the measurement window into pre-disclosure (July–December of year t) and post-disclosure (May–December of year $t + 1$) sub-periods, we find that SC-RDI significantly predicts comovement in the post-disclosure window ($\beta = +0.016$, $t = 2.96$), establishing that the relationship is not driven by reverse causality from returns to disclosure. The relationship also appears in the pre-disclosure window ($t = 2.71$), suggesting that supply chain information reaches markets through channels beyond the formal annual report.

Third, not all supply chain risk narratives are priced equally. Disruption risk disclosure is the dominant driver of comovement ($\beta = +0.123$, $t = 3.15$), consistent with the theoretical prediction that risk dimensions revealing *shared* exposure to common shocks should be the strongest predictors of coordinated return movement. Trade dependence risk also enters significantly ($\beta = +0.035$, $t = 2.98$). Concentration, input price, and logistics risks are not consistently significant, underscoring the methodological importance of dictionary disaggregation.

Fourth, SC-RDI functions as a transparency signal rather than a risk warning. Higher disclosure predicts a 23% lower probability of material operating disruption (OR=0.772, $z = -5.12$), consistent with well-governed firms both disclosing more and experiencing fewer operational shocks.

Fifth, the mechanism through which disclosure translates into comovement appears to operate through institutional investor attention. The SC-RDI coefficient is significantly larger in firms with high institutional ownership ($\beta = +0.027$, $t = 4.20$) than in firms with low institutional ownership ($\beta = +0.008$, $t = 1.44$; difference $t = 2.11$), suggesting that institutional investors are primary processors—rather than substitutes for—textual supply chain information. Neither analyst coverage nor institutional ownership mediates the effect, redirecting attention toward direct trading by sophisticated investors as the likely pricing channel.

Our findings carry implications for disclosure regulation, investment practice, and future research. For regulators, the extreme sparsity of supply chain risk disclosure (73% of firm-years record zero risk language) suggests that current MD&A requirements do not elicit meaningful narratives about what is arguably the most consequential operational risk firms face. For investors, textual analysis of MD&A sections—particularly disruption-related language—can inform portfolio risk assessment in a market where the vast majority of supply chain partners (98%) lack observable stock prices. For researchers, our results demonstrate that (i) the choice of return proxy for unobservable partners can determine whether a disclosure–return relationship is detected at all; (ii) dictionary disaggregation is essential for identifying which content dimensions drive observed effects; and (iii) the post-disclosure window decomposition provides a feasible temporal falsification test for annual-report-based disclosure studies. More broadly, our findings connect two rapidly growing but largely separate literatures: the text-based measurement of supply chain risk exposure [Wu \(2024\)](#); [Kroes et al. \(2025\)](#) and the network-based asset pricing of production linkages [Ramírez \(2024\)](#); [Yang and Zhao \(2023\)](#). Integrating textual disclosure measures into formal network asset pricing models—for instance, allowing the perceived strength of network linkages to depend on the intensity of qualitative risk disclosure—represents a promising direction for future theoretical work.

Three institutional features of our setting warrant caution in generalizing our findings. First, China's MD&A framework grants substantial managerial discretion, producing the extreme disclosure sparsity that underpins our transparency-signal result; in jurisdictions with more prescriptive disclosure requirements (e.g., the United States),

the signal-to-noise ratio of textual supply chain measures may differ. Second, the predominance of non-listed supply chain partners (98%) makes our characteristic-matched portfolio innovation particularly consequential for the Chinese setting; in markets with more listed small and medium enterprises, researchers may rely less on proxy returns, and the comovement measure will contain a larger share of true pairwise correlations. Third, China's retail-investor-dominated market structure may amplify sentiment-based comovement relative to markets with greater institutional participation, though our heterogeneity results suggest that the institutional channel is operative even within China's retail-heavy environment. Replicating the disclosure–comovement relationship in other institutional contexts—particularly in developed markets with different disclosure regimes and supply chain structures—is an important avenue for future research.

A. Supply Chain Risk Dictionary

Table 12
Complete Dictionary (37 Keywords in 5 Categories)

Category	Chinese Keywords	English Translation
Concentration	客户集中 供应商集中 依赖风险 单一供应商 核心客户依赖 集中度较高 大客户依赖	Customer concentration Supplier concentration Dependence risk Single supplier Key customer dependence High concentration Major customer dependence
Disruption	断供 供应中断 产能不足 交付延迟 物流中断 停产风险 原材料短缺 供应稳定性 复工延迟	Supply cut Supply interruption Capacity shortage Delivery delay Logistics disruption Production halt risk Raw material shortage Supply stability Resumption delay
Input Price	原材料价格波动 采购成本上升 原料价格上涨 成本传导 议价能力不足 原材料成本 价格波动风险	Raw material price fluctuation Procurement cost increase Raw material price rise Cost pass-through Weak bargaining power Raw material cost Price fluctuation risk
Trade Dependence	进口依赖 出口依赖 贸易摩擦 关税风险 地缘政治 技术封锁 海外依赖 外部制裁	Import dependence Export dependence Trade friction Tariff risk Geopolitics Technology blockade Overseas dependence External sanctions
Logistics/Operational	物流成本 运输中断 库存压力 产能瓶颈 渠道风险 外包风险 核心零部件	Logistics cost Transportation disruption Inventory pressure Capacity bottleneck Channel risk Outsourcing risk Core components

B. Mediation Analysis Details

Table 13

Mediation Analysis: Analyst Coverage and Institutional Ownership

Path	Analyst		InsInvestorProp	
	Coefficient	<i>t/z</i>	Coefficient	<i>t/z</i>
Path <i>a</i> : SC-RDI → Mediator	+0.000	+0.19	+0.000	+0.27
Path <i>b</i> : Mediator → Comovement	+0.000	+1.18	−0.001***	−8.81
Path <i>c</i> (total effect)	+0.017***	+3.82	+0.017***	+3.82
Path <i>c'</i> (direct effect)	+0.017***	+3.82	+0.017***	+3.82
Indirect effect ($a \times b$)	+0.000		−0.000	
Sobel <i>z</i>	+0.187	(<i>p</i> = 0.852)	−0.269	(<i>p</i> = 0.788)
% Mediated	0.0%		−0.0%	

Notes. Baron–Kenny three-step mediation with Sobel–Goodman tests. All regressions include full controls and industry×year FE.

C. V1 Results: Industry-Portfolio-Based Comovement

This appendix reports results from our initial specification (V1), in which non-listed partner returns were proxied using equal-weighted industry portfolio returns rather than size-BM characteristic-matched portfolios. The industry-portfolio approach creates a mechanical correlation between Comovement and industry×year fixed effects, as both are driven by industry-level return variation. As Table 14 shows, the V1 specification yields a null baseline coefficient ($\beta = -0.0002$, $t = -0.06$), in sharp contrast to the V2 result ($\beta = +0.017$, $t = 3.82$). This comparison underscores the methodological importance of the return proxy choice.

Table 14

V1 Baseline: Industry-Portfolio Comovement

	V1 (Industry Portfolio Y)	V2 (Size-BM Portfolio Y)
ln_SC_RDI_cont (Ind×Year FE)	−0.0002 (−0.06)	+0.017*** (+3.82)
<i>N</i>	45,391	45,391
<i>R</i> ²	0.179	0.229

The V1 five-category decomposition (Table 15) also reveals differences: trade dependence and logistics risks were insignificant or negative in V1, but become positive and significant in V2.

Table 15

V1 Five-Category Decomposition

Category	V1 β (<i>t</i>)	V2 β (<i>t</i>)
Concentration	−0.009 (−0.57)	−0.011 (−0.71)
Disruption	+0.120*** (+3.30)	+0.123*** (+3.15)
Input Price	−0.008 (−1.11)	+0.008 (+1.13)
Trade Dependence	+0.017 (+1.45)	+0.035*** (+2.98)
Logistics	−0.032** (−2.29)	+0.022* (+1.67)

Table 16 compares the V1 and V2 robustness results. In V1, only 1 of 18 specifications (Firm FE) was significant; in V2, 14 of 18 are significant at the 5% level or better.

Table 16

V1 vs. V2 Robustness Comparison

Test	V1 β (<i>t</i>)	V2 β (<i>t</i>)
Baseline (Ind×Year FE)	−0.0002 (−0.05)	+0.017*** (+3.82)
PSM	−0.002 (−0.43)	+0.017*** (+3.18)
Fama–MacBeth	−0.004 (−0.82)	+0.010** (+2.21)
Alt X (raw)	−0.001 (−0.22)	+0.008*** (+2.93)
Alt X (dummy)	−0.0002 (−0.06)	+0.012*** (+4.32)
Manufacturing	−0.005 (−1.00)	+0.015*** (+3.12)
Non-SOE	+0.001 (+0.26)	+0.015*** (+3.02)
SOE	−0.009 (−0.97)	+0.017* (+1.88)
Firm FE	+0.008*** (+2.75)	+0.010*** (+3.29)
Heckman	−0.0003 (−0.06)	+0.017*** (+3.82)
Significant (5%)	1 / 18	14 / 18

D. COVID-19 DID Analysis

We test whether the COVID-19 pandemic differentially affected comovement for firms with higher pre-pandemic SC-RDI. Treatment firms are those with SC-RDI > 0 in 2018–2019; the post period is 2020–2021. Table 17 reports DID estimates.

Table 17
COVID-19 Difference-in-Differences

Specification	β (Treat $\times$ Post)	t	N
Main DID	−0.001	−0.17	14,085
Placebo (fake 2018)	+0.013	+1.42	12,137
Alt treatment (median split)	−0.001	−0.17	14,085
Narrow window (2019 vs 2020)	−0.019	−1.42	6,909

Notes. All specifications include full controls and industry FE. Firm-clustered standard errors. The placebo test uses 2018 as a fake treatment year with data from 2016–2019. The insignificant placebo coefficient supports the parallel trends assumption.

The DID coefficient is not significant ($\beta = -0.001$, $t = -0.17$), and the placebo test produces an insignificant coefficient ($\beta = +0.013$, $t = 1.42$). These results suggest that the SC-RDI–comovement relationship is not driven by the COVID-19 shock but is a more general phenomenon.

E. Falsification Tests: Placebo Y and Placebo X

This appendix reports the two falsification tests described in Section 7.2.

Placebo Y: Random non-partner comovement. For each firm-year, we randomly select five firms from the same one-digit industry that are *not* in the focal firm's disclosed supply chain, compute pairwise return correlations, and average them to form a placebo comovement measure ($Comovement^{placebo}$). We then regress this placebo comovement on SC-RDI using the baseline specification. If SC-RDI captures supply-chain-specific information, it should not predict comovement with randomly assigned non-partners. Table 18 Panel A reports the results. The coefficient on SC-RDI when predicting placebo comovement is substantially smaller than when predicting real supply chain comovement in the same estimation sample, and the difference between the real and placebo coefficients is economically meaningful.

Placebo X: General disclosure proxies. We add three general disclosure quality proxies to the baseline regression: MD&A sentence count (a measure of disclosure volume), firm age, and ROA (proxies for disclosure maturity and quality). If SC-RDI captures only generic disclosure verbosity, its coefficient should attenuate toward zero when these proxies are included. Table 18 Panel B reports the results. SC-RDI remains statistically significant, while the general disclosure proxies are not individually or jointly significant predictors of comovement. This confirms that SC-RDI captures supply-chain-specific content rather than general disclosure quantity.

Table 18
Falsification Tests

<i>Panel A: Placebo Y — Random Non-Partner Comovement</i>			
Dependent Variable	β (SC-RDI)	t	N
Real Comovement (supply chain partners)	+0.0163***	+3.82	45,355
Placebo Comovement (random non-partners)	+0.0048	+1.54	45,355
Difference (Real – Placebo): +0.0114, $t = 2.28^{**}$			
<i>Panel B: Placebo X — General Disclosure Proxies</i>			
Variable	β	t	p
ln_SC_RDI_cont (baseline, same sample)	+0.0162***	+3.79	< 0.01
n_mda_sent_ln (standalone)	+0.0018	+1.73	0.08
FirmAge (standalone)	−0.0006	−2.65	0.01
ln_SC_RDI_cont (joint with all proxies)	+0.0149***	+3.43	< 0.01

Notes. Panel A: Placebo comovement computed from 5 randomly selected same-industry non-partners per firm-year. Regression includes full controls and industry×year FE. The placebo coefficient is insignificant ($p = 0.12$) and the real-placebo difference is significant, supporting supply-chain specificity. Panel B: Each general disclosure proxy is tested both individually and jointly with SC-RDI. SC-RDI remains significant in all specifications, while the general disclosure proxies do not consistently predict comovement. ***, **, * denote 1%, 5%, 10% significance.

References

- Acemoglu, D., Carvalho, V. M., Ozdaglar, A., & Tahbaz-Salehi, A. (2012). The network origins of aggregate fluctuations. *Econometrica*, 80(5), 1977–2016.
- Akerlof, G. A. (1970). The market for “lemons”: Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, 84(3), 488–500.
- Ahern, K. R., & Harford, J. (2014). The importance of industry links in merger waves. *Journal of Finance*, 69(2), 527–576.
- Barberis, N., Shleifer, A., & Wurgler, J. (2005). Comovement. *Journal of Financial Economics*, 75(2), 283–317.
- Barrot, J.-N., & Sauvagnat, J. (2016). Input specificity and the propagation of idiosyncratic shocks in production networks. *Quarterly Journal of Economics*, 131(3), 1543–1592.
- Cen, L., Maydew, E. L., Zhang, L., & Zuo, L. (2021). Customer–supplier relationships and corporate disclosure. *Journal of Accounting Research*, 59(2), 525–571.
- Cohen, L., & Frazzini, A. (2008). Economic links and predictable returns. *Journal of Finance*, 63(4), 1977–2011.
- Dye, R. A. (1985). Disclosure of nonproprietary information. *Journal of Accounting Research*, 23(1), 123–145.
- Goldstein, I., Spatt, C. S., & Ye, M. (2021). Big data in finance. *Review of Financial Studies*, 34(7), 3213–3225.
- Hassan, T. A., Hollander, S., van Lent, L., & Tahoun, A. (2019). Firm-level political risk: Measurement and effects. *Quarterly Journal of Economics*, 134(4), 2135–2202.
- Hoberg, G., & Phillips, G. (2016). Text-based network industries and endogenous product differentiation. *Journal of Political Economy*, 124(5), 1423–1465.
- Hong, H., & Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6), 2143–2184.
- Li, Z., & Yang, J. (2019). Beyond the numbers: The information content of MD&A textual disclosure in China. *China Journal of Accounting Research*, 12(3), 307–328.
- Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1), 35–65.
- Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies*, 22(1), 435–480.
- Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8), 2586–2624.
- Borusyak, K., Hull, P., & Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1), 181–213.
- Engelberg, J. E., Reed, A. V., & Ringgenberg, M. C. (2012). How are shorts informed? Short sellers, news, and information processing. *Journal of Financial Economics*, 105(2), 260–278.
- Cohen, L., & Lou, D. (2012). Complicated firms. *Journal of Financial Economics*, 104(2), 383–400.
- Campbell, J. Y., Lettau, M., Malkiel, B. G., & Xu, Y. (2001). Have individual stocks become more volatile? An empirical exploration of idiosyncratic risk. *Journal of Finance*, 56(1), 1–43.
- Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3), 1139–1168.
- Gow, I. D., Larcker, D. F., & Reiss, P. C. (2016). Causal inference in accounting research. *Journal of Accounting Research*, 54(2), 477–523.
- Jegadeesh, N., & Wu, D. (2013). Word power: A new approach for content analysis. *Journal of Financial Economics*, 110(3), 712–729.
- Piotroski, J. D., Wong, T. J., & Zhang, T. (2015). Political incentives to suppress negative information: Evidence from Chinese listed firms. *Journal of Accounting Research*, 53(2), 405–459.
- Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics*, 87(3), 355–374.
- Verrecchia, R. E. (1983). Discretionary disclosure. *Journal of Accounting and Economics*, 5, 179–194.
- Baiman, S., & Rajan, M. (2002). The role of information and opportunism in the choice of buyer–supplier relationships. *Journal of Accounting Research*, 40(2), 247–278.
- Bai, H., Liu, M., Wan, C., & Yuksel, H. Z. (2024). Narrative risk disclosures: Implications for systematic risk exposures and asset prices. Working Paper, Columbia Business School.
- Cen, L., Chen, F., Hou, Y., & Richardson, G. D. (2018). Strategic disclosures of litigation loss contingencies when customer–supplier relationships are at risk. *The Accounting Review*, 93(2), 137–159.
- Costello, A. M., Levy, H., & Nikolaev, V. V. (2024). Uncovering information: Can AI tell us where to look? NBER Working Paper No. w210937.
- Davis, A. K., Piger, J. M., & Sedor, L. M. (2012). Beyond the numbers: Measuring the information content of earnings press release language. *Contemporary Accounting Research*, 29(3), 845–868.
- Duan, B., Wang, R., & Zhang, R. (2022). Economic links and stock returns in Chinese A-share market. *Journal of Financial Research*, 500(2), 171–188.
- Kroes, J. R., Manikas, A. S., Baig, A. S., & Land, M. J. (2025). Supply chain risk disclosures in 10-K filings: Implications for inventory slack and stock returns in unstable demand markets. *International Journal of Production Research*, 63(16), 6089–6107.
- Ramírez, C. A. (2024). Firm networks and asset returns. *Review of Financial Studies*, 37(10), 3050–3091.
- Wu, D. A. (2024). Text-based measure of supply chain risk exposure. *Management Science*, 70(7), 4781–4801.
- Yang, C., & Zhao, Y. (2023). Supply chains and risk premia in Chinese stock market: A sorted-portfolio approach. *International Studies of Economics*, 18(3), 277–305.